SG No. 174 C_{3h}^{1} $P\bar{6}$ [hexagonal]

* plus set: $+[0, 0, 0]$
* Wyckoff site: `1a`, site symmetry: `-6..`

Table 1: Wyckoff bond: `1a@1a`, bond symmetry: `-6..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, 0]$	`[1,2,3,-4,-5,-6]`

Table 2: Wyckoff bond: `3b@1a`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, 0]$	`[1,4]`
2	$[-Y, X-Y, 0]$	$[0, 0, 0]$	`[2,5]`
3	$[-X+Y, -X, 0]$	$[0, 0, 0]$	`[3,6]`

Table 3: Wyckoff bond: `6c@1a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, 0]$	`[1]`
2	$[-Y, X-Y, Z]$	$[0, 0, 0]$	`[2]`
3	$[-X+Y, -X, Z]$	$[0, 0, 0]$	`[3]`
4	$[X, Y, -Z]$	$[0, 0, 0]$	`[4]`
5	$[-Y, X-Y, -Z]$	$[0, 0, 0]$	`[5]`
6	$[-X+Y, -X, -Z]$	$[0, 0, 0]$	`[6]`

* Wyckoff site: `1b`, site symmetry: `-6..`

Table 4: Wyckoff bond: `1a@1b`, bond symmetry: `-6..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,-4,-5,-6]`

Table 5: Wyckoff bond: `3b@1b`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,4]`
2	$[-Y, X-Y, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[2,5]`
3	$[-X+Y, -X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[3,6]`

Table 6: Wyckoff bond: `6c@1b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1]`
2	$[-Y, X-Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[2]`

continued ...

Table 6

No.	vector	center	mapping
3	$[-X+Y,\ -X,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[3]`
4	$[X,\ Y,\ -Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[4]`
5	$[-Y,\ X-Y,\ -Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[5]`
6	$[-X+Y,\ -X,\ -Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[6]`

* Wyckoff site: `1c`, site symmetry: `-6..`

Table 7: Wyckoff bond: `1a@1c`, bond symmetry: `-6..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[1,2,3,-4,-5,-6]`

Table 8: Wyckoff bond: `3b@1c`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[1,4]`
2	$[-Y,\ X-Y,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[2,5]`
3	$[-X+Y,\ -X,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[3,6]`

Table 9: Wyckoff bond: `6c@1c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[3]`
4	$[X,\ Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[4]`
5	$[-Y,\ X-Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[5]`
6	$[-X+Y,\ -X,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[6]`

* Wyckoff site: `1d`, site symmetry: `-6..`

Table 10: Wyckoff bond: `1a@1d`, bond symmetry: `-6..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[1,2,3,-4,-5,-6]`

Table 11: Wyckoff bond: `3b@1d`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[1,4]`
2	$[-Y,\ X-Y,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[2,5]`
3	$[-X+Y,\ -X,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[3,6]`

Table 12: Wyckoff bond: `6c@1d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[1]`
2	$[-Y, X-Y, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[2]`
3	$[-X+Y, -X, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[3]`
4	$[X, Y, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[4]`
5	$[-Y, X-Y, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[5]`
6	$[-X+Y, -X, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[6]`

* Wyckoff site: `1e`, site symmetry: `-6..`

Table 13: Wyckoff bond: `1a@1e`, bond symmetry: `-6..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[1,2,3,-4,-5,-6]`

Table 14: Wyckoff bond: `3b@1e`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[1,4]`
2	$[-Y, X-Y, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[2,5]`
3	$[-X+Y, -X, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[3,6]`

Table 15: Wyckoff bond: `6c@1e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[1]`
2	$[-Y, X-Y, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[2]`
3	$[-X+Y, -X, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[3]`
4	$[X, Y, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[4]`
5	$[-Y, X-Y, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[5]`
6	$[-X+Y, -X, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[6]`

* Wyckoff site: `1f`, site symmetry: `-6..`

Table 16: Wyckoff bond: `1a@1f`, bond symmetry: `-6..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	`[1,2,3,-4,-5,-6]`

Table 17: Wyckoff bond: `3b@1f`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	`[1,4]`
2	$[-Y, X-Y, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	`[2,5]`

continued ...

Table 17

No.	vector	center	mapping
3	$[-X+Y,\ -X,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	[3,6]

Table 18: Wyckoff bond: 6c@1f, bond symmetry: 1

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	[1]
2	$[-Y,\ X-Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	[2]
3	$[-X+Y,\ -X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	[3]
4	$[X,\ Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	[4]
5	$[-Y,\ X-Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	[5]
6	$[-X+Y,\ -X,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	[6]

* Wyckoff site: 2g, site symmetry: 3..

Table 19: Wyckoff bond: 2a@2g, bond symmetry: 3..

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$[0,\ 0,\ z]$	[1,2,3]
2	$[0,\ 0,\ -Z]$	$[0,\ 0,\ -z]$	[4,5,6]

Table 20: Wyckoff bond: 6b@2g, bond symmetry: 1

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[0,\ 0,\ z]$	[1]
2	$[-Y,\ X-Y,\ Z]$	$[0,\ 0,\ z]$	[2]
3	$[-X+Y,\ -X,\ Z]$	$[0,\ 0,\ z]$	[3]
4	$[X,\ Y,\ -Z]$	$[0,\ 0,\ -z]$	[4]
5	$[-Y,\ X-Y,\ -Z]$	$[0,\ 0,\ -z]$	[5]
6	$[-X+Y,\ -X,\ -Z]$	$[0,\ 0,\ -z]$	[6]

* Wyckoff site: 2h, site symmetry: 3..

Table 21: Wyckoff bond: 2a@2h, bond symmetry: 3..

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	[1,2,3]
2	$[0,\ 0,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$	[4,5,6]

Table 22: Wyckoff bond: 6b@2h, bond symmetry: 1

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	[1]
2	$[-Y,\ X-Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	[2]

continued ...

Table 22

No.	vector	center	mapping
3	$[-X+Y, -X, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	[3]
4	$[X, Y, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	[4]
5	$[-Y, X-Y, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	[5]
6	$[-X+Y, -X, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	[6]

* Wyckoff site: 2i, site symmetry: 3..

Table 23: Wyckoff bond: 2a@2i, bond symmetry: 3..

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	[1,2,3]
2	$[0, 0, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	[4,5,6]

Table 24: Wyckoff bond: 6b@2i, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	[1]
2	$[-Y, X-Y, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	[2]
3	$[-X+Y, -X, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	[3]
4	$[X, Y, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	[4]
5	$[-Y, X-Y, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	[5]
6	$[-X+Y, -X, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	[6]

* Wyckoff site: 3j, site symmetry: m..

Table 25: Wyckoff bond: 3a@3j, bond symmetry: m..

No.	vector	center	mapping
1	$[X, Y, 0]$	$[x, y, 0]$	[1,4]
2	$[-Y, X-Y, 0]$	$[-y, x-y, 0]$	[2,5]
3	$[-X+Y, -X, 0]$	$[-x+y, -x, 0]$	[3,6]

Table 26: Wyckoff bond: 3b@3j, bond symmetry: m..

No.	vector	center	mapping
1	$[0, 0, Z]$	$[x, y, 0]$	[1,-4]
2	$[0, 0, Z]$	$[-y, x-y, 0]$	[2,-5]
3	$[0, 0, Z]$	$[-x+y, -x, 0]$	[3,-6]

Table 27: Wyckoff bond: 6c@3j, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, 0]$	[1]

continued ...

Table 27

No.	vector	center	mapping
2	$[-Y,\ X-Y,\ Z]$	$[-y,\ x-y,\ 0]$	[2]
3	$[-X+Y,\ -X,\ Z]$	$[-x+y,\ -x,\ 0]$	[3]
4	$[X,\ Y,\ -Z]$	$[x,\ y,\ 0]$	[4]
5	$[-Y,\ X-Y,\ -Z]$	$[-y,\ x-y,\ 0]$	[5]
6	$[-X+Y,\ -X,\ -Z]$	$[-x+y,\ -x,\ 0]$	[6]

* Wyckoff site: 3k, site symmetry: m..

Table 28: Wyckoff bond: 3a@3k, bond symmetry: m..

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[x,\ y,\ \frac{1}{2}\right]$	[1,4]
2	$[-Y,\ X-Y,\ 0]$	$\left[-y,\ x-y,\ \frac{1}{2}\right]$	[2,5]
3	$[-X+Y,\ -X,\ 0]$	$\left[-x+y,\ -x,\ \frac{1}{2}\right]$	[3,6]

Table 29: Wyckoff bond: 3b@3k, bond symmetry: m..

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[x,\ y,\ \frac{1}{2}\right]$	[1,-4]
2	$[0,\ 0,\ Z]$	$\left[-y,\ x-y,\ \frac{1}{2}\right]$	[2,-5]
3	$[0,\ 0,\ Z]$	$\left[-x+y,\ -x,\ \frac{1}{2}\right]$	[3,-6]

Table 30: Wyckoff bond: 6c@3k, bond symmetry: 1

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[x,\ y,\ \frac{1}{2}\right]$	[1]
2	$[-Y,\ X-Y,\ Z]$	$\left[-y,\ x-y,\ \frac{1}{2}\right]$	[2]
3	$[-X+Y,\ -X,\ Z]$	$\left[-x+y,\ -x,\ \frac{1}{2}\right]$	[3]
4	$[X,\ Y,\ -Z]$	$\left[x,\ y,\ \frac{1}{2}\right]$	[4]
5	$[-Y,\ X-Y,\ -Z]$	$\left[-y,\ x-y,\ \frac{1}{2}\right]$	[5]
6	$[-X+Y,\ -X,\ -Z]$	$\left[-x+y,\ -x,\ \frac{1}{2}\right]$	[6]

* Wyckoff site: 6l, site symmetry: 1

Table 31: Wyckoff bond: 6a@6l, bond symmetry: 1

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[x,\ y,\ z]$	[1]
2	$[-Y,\ X-Y,\ Z]$	$[-y,\ x-y,\ z]$	[2]
3	$[-X+Y,\ -X,\ Z]$	$[-x+y,\ -x,\ z]$	[3]
4	$[X,\ Y,\ -Z]$	$[x,\ y,\ -z]$	[4]
5	$[-Y,\ X-Y,\ -Z]$	$[-y,\ x-y,\ -z]$	[5]
6	$[-X+Y,\ -X,\ -Z]$	$[-x+y,\ -x,\ -z]$	[6]